CONFIDENCE FALLBACK: SELECTIVE STOPPING FOR CONTINUOUS LATENT REASONING

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ABSTRACT

Continuous latent reasoning searches in hidden space, but deployment still requires deciding how many latent steps to allocate and whether to trust the resulting prediction. We study **Confidence Fallback (CF)**, an inference policy on frozen Coconut weights that routes each instance to a structure-derived budget, scores the aligned prediction, and refines only a low-confidence tail. On a 419-question ProsQA development set, CF reaches 93.89±0.76% over five inference seeds (best 95.23%) while refining 7.2%, gaining +10.1 pp over Fixed 3 on average (+11.4 pp best seed). A development-selected gate adds 2.08 pp across 53 shifted ProsQA slices. Five external pilots give mixed evidence for the route-then-gate skeleton, not one task-independent controller. CF adds no decoded thought tokens relative to the frozen base; commercial direct-answer systems use 25–82× more reported tokens, although this reference is not scale- or compute-matched. Code: https://github.com/chuhan-0215/confidence_fallback

Keywords: adaptive stopping; continuous latent reasoning; confidence fallback; Coconut; ProsQA

1 INTRODUCTION

Chain-of-thought prompting elicits explicit reasoning traces from language models [3]. Coconut instead feeds the last hidden state back as the next input embedding, so each latent step can maintain a superposition over multiple search frontiers [1; 2] and outperforms discrete chains on ProsQA graph reachability [1]. At inference time, however, the model still needs an instance-specific latent budget and a rule for accepting or revising the resulting prediction.

Prior stopping rules treat these choices separately. A uniform three-step budget reaches 83.8% on our ProsQA split even though solvable depth correlates with graph structure (r=0.543). Depth-aware routing removes most of this mismatch (**Depth** 93.75%, **Auto BFS** 93.1%). The structure-routed and universal-refinement paths make 22 and 31 errors, respectively, with only 17 shared errors. Refining every instance reaches only 92.6%, below routing alone; this complementarity motivates a sequential policy that first aligns depth and then decides whether the aligned prediction is trustworthy.

We propose **Confidence Fallback (CF)**, which first routes each instance to a structure-derived budget and then invokes neighborhood refinement only below a development-selected confidence threshold. On the 419-question ProsQA development set, CF reaches 93.89±0.76% over five inference seeds (best 95.23%) while refining 7.2%. We state a sufficient-condition rationale for gating after routing (Definition 1, Proposition 1, Lemma 1), report the evaluated accuracy–compute trade-off, and examine shifted and external pilots without claiming a frozen task-independent controller.

Related work. Coconut [1] and the superposition analysis of Zhu et al. [2] improve how reasoning is represented in latent space; CF is complementary and operates purely at inference time on frozen weights. Verifiers such as Cobbe et al. [4] score explicit solution traces, whereas CF scores a continuous hidden state at the routed depth and applies refinement only to a 7.2% tail.

2 PROBLEM FORMULATION

In ProsQA graph reachability [1], each instance x is a rooted directed graph with two candidate targets; y is the reachability label and d the BFS depth from the root. Coconut generates n continuous thoughts and emits a prediction $\hat{y}$. A *policy* chooses n per instance and whether to accept $\hat{y}$. Throughout, we clamp routed depth to $[n_{\min}, n_{\max}]$ with $n_{\min}$=3 and $n_{\max}$=8, matching the deployment budget used in our experiments.

Definition 1 (Two-stage stopping errors)**.** For analysis, we operationally categorize deployment failures under a depth-selecting policy into two dominant (non-exclusive) modes: (i) *depth mismatch* E_d: the budget stops before (or wanders past) the depth at which the instance is solvable; and (ii) *commitment error* E_c: at an aligned depth, the model commits to a prediction whose stop confidence is unreliable; $E \approx E_d \cup E_c$ is an operational approximation, not a strict partition.

3 CONFIDENCE FALLBACK

CF repairs the two error types *in that order*: routing aligns the latent budget with instance structure, and confidence gating decides whether the aligned prediction is trustworthy. **Route and score.** At each latent step n, we form stability features $s_n = (b_n, r_n, c_n)$: answer bucket b_n, repeat streak r_n, and change flag $c_n = \mathbf{1}[\hat{y}_n \neq \hat{y}_{n-1}]$. One forward pass to the structure-routed depth n_0 yields $\hat{y}_0$, h_{n_0}, and s_0, which the stop head scores:

$$n_0 = \max\big(n_{\min},\ \min(d,\ n_{\max})\big), \qquad p_0 = \sigma\big(g(h_{n_0}, n_0, s_0)\big). \tag{1}$$

Gate. The policy commits to $\hat{y}_0$ when $p_0 \geq \tau_{\text{fb}}$. Otherwise it evaluates candidate depths from a floor n_{fl} up to $n_{\max}$ and stops at the first candidate that reaches the separately calibrated online-stop threshold τ_{stop}. Define

$$n^\star = \min(\{n \in \{n_{\text{fl}}, \ldots, n_{\max}\} : p_n \geq \tau_{\text{stop}}\} \cup \{n_{\max}\}), \quad \hat{y} = \begin{cases} \hat{y}_0, & p_0 \geq \tau_{\text{fb}}, \\ \hat{y}_{n^\star}, & p_0 < \tau_{\text{fb}}. \end{cases} \tag{2}$$

Routing targets E_d and gated refinement targets the residual E_c (Definition 1). In our implementation, n_{fl}=3, $n_{\max}$=8, and each evaluated candidate requires a separate model invocation; if none passes τ_{stop}, the policy returns the $n_{\max}$ candidate.

Proposition 1 (Route-before-gate)**.** *Under A1–A3 in Lemma 1, scoring after depth alignment removes depth-mismatch mass from the gate input. Gate-first instead pools depth and commitment risk, so one threshold cannot generally reproduce route-then-gate behavior.*

Lemma 1 (Confidence contamination under overlapping error modes)**.** *Assume* (A1) *structure routing removes depth-mismatch mass without harming aligned instances,* (A2) *on aligned depth, g ranks commitment risk above chance, and* (A3) *the confidence distributions of depth-mismatch and commitment-error instances overlap. If p_0 is instead scored while $E_d > 0$, then* $\Pr(\text{wrong} \mid p_0 < \tau)$ *contains depth-misallocation mass, and a single threshold cannot in general cleanly isolate the two error modes.*

Proof sketch. Routing removes E_d before scoring (A1–A2); gate-first pools overlapping modes (A3), so no single τ cleanly separates them (Figure 2B).

Algorithm 1 Confidence Fallback

Input: x, τ_{fb}, τ_{stop}; Coconut f; stop head g
Output: Answer $\hat{y}$

$n_0 \leftarrow \max\big(n_{\min}, \min(\textsc{BfsDepth}(x), n_{\max})\big)$
$(\hat{y}_0, h_{n_0}) \leftarrow f(x, n_0)$
$s_0 \leftarrow \textsc{StabFeat}(\hat{y}_0, n_0)$
$p_0 \leftarrow \sigma\big(g(h_{n_0}, n_0, s_0)\big)$
if $p_0 \geq \tau_{\text{fb}}$ **then**
  return $\hat{y}_0$
else
  return $\textsc{Refine}(x, n_{\text{fl}}, n_{\max}, \tau_{\text{stop}})$
end if

Algorithm 2 Neighborhood Refinement

Input: x; f; g; floor n_{fl}; cap $n_{\max}$; τ_{stop}
Output: Answer $\hat{y}$

for $n = n_{\text{fl}}, \ldots, n_{\max}$ **do**
  $(\hat{y}, h_n) \leftarrow f(x, n)$
  $s_n \leftarrow \textsc{StabFeat}(\hat{y}, n)$
  $p \leftarrow \sigma\big(g(h_n, n, s_n)\big)$
  if $p \geq \tau_{\text{stop}}$ **then**
    break
  end if
end for
return $\hat{y}$

Stop head. A two-layer, 106K-parameter MLP maps $h_n \in \mathbb{R}^{768}$, step n, and s_n to a stop logit. It uses per-step *current-correctness* labels: 1,608 training examples from 201 questions (1,082 positive; 67.3%) and 400 per-step validation examples from 50 questions, with focal BCE [5] (γ=2). We calibrate τ_{stop} on those 50 questions and select τ_{fb}=0.48 on the 419-question development set (Algorithms 1–2).

Table 1: **ProsQA direct-answer reference (not scale/compute matched).** Three APIs use a random 100-question subset (seed 42); GPT-4.1 variants use all 419. API latency includes network I/O; two Qwen API failures are counted as incorrect.

Model	Type	Acc.(%)	Avg.Tokens	Latency(s)
CF (GPT-2; 5-seed mean)	**Latent reasoning**	**93.9**	**14.2**	**–**
Qwen3.7-Max	Flagship direct	94.0	1163.9	26.06
GPT-5.4	Flagship direct	87.0	480.9	2.48
Kimi-K2.6	Direct-answer API	77.0	484.8	4.53
GPT-4.1	Flagship direct	65.6	355.0	2.30
GPT-4.1-Mini	Lightweight direct	53.5	354.2	1.74

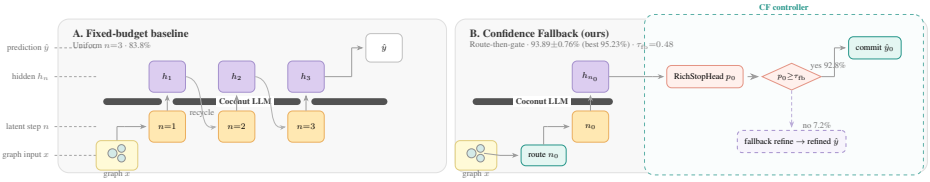


Figure 1: **CF inference controller.** **A**, Fixed three-step budget (83.8%). **B**, CF routes n_0 from $\textsc{BfsDepth}(x)$, scores $p_0{=}g(h_{n_0}, n_0, s_0)$, commits on 92.8% of instances when $p_0 \geq \tau_{\text{fb}}$, and refines the remaining 7.2% ($\bar{n}{\approx}3.51$).

4 Experiments

4.1 Experimental Setup

Model and hardware. All policies share the frozen GPT-2-based Coconut step-300 checkpoint [1] and stop head. ProsQA runs use one NVIDIA A100. **Data and separation.** Of 419 questions, 201/50 supply stop-head training/validation examples and 168 were initially held out. Later policy selection, including τ_{fb}, uses all 419, so full-set numbers are development results, not unbiased test estimates. The 53 derived slices also informed guard design. **Baselines.** With no per-instance oracle sweep and $\bar{n} \leq 4.5$, **Fixed 3** uses three steps; **Auto BFS** and **Depth** use BFS depth; **Universal** always searches; CF uses $\tau_{\text{fb}}{=}0.48$ and $n_{\text{fl}}{=}3$. **Metrics.** We report accuracy, gain, refinement rate, selected latent depth, and available latency.

4.2 Main Results and Analysis

Figure 2A summarizes the five-policy ablation on the development set. Depth-aware routing recovers most of the fixed-budget gap: **Depth** reaches 93.75% (+10.0 pp) and **Auto BFS** reaches 93.1% (+9.3 pp) versus **Fixed 3** at 83.8%. Universal refinement (**Universal**) reaches 92.6% (+8.8 pp) at substantially higher compute, confirming that blind refinement cannot replace routing. Across five inference seeds, CF reaches 93.89±0.76% (mean±standard deviation; best 95.23%, seed 99), +10.1 pp over **Fixed 3** for the mean (+11.4 pp at the best seed). It refines 7.2% of instances and selects $\bar{n}{=}3.51$ latent steps, 1.17× the Fixed 3 budget (Figure 2B).

Commercial reference. Table 1 is a decoding reference, not a controlled capability or FLOP comparison: samples, scale, serving, and accounting differ, and token ratios exclude latent compute.

Error decomposition. Of 419 instances, refinement recovers 5 routing-wrong cases, misleads 14 routing-correct ones, and leaves 17 jointly wrong, giving a 95.94% path-union oracle ceiling (Figure 2C). Instance 3 has $p_0{=}0.47 < \tau_{\text{fb}}$ and triggers refinement but remains wrong; instance 111 has $p_0{=}0.51 \geq \tau_{\text{fb}}$ although a correct refinement path exists. **Compute profile.** CF decodes no thought tokens ($\approx$ 14.2 reported tokens/question vs. 49.4 for explicit CoT [1]); its selected latent depth is $\bar{n}{=}3.51$, 1.17× Fixed 3. The evaluator materializes candidate depths for analysis, so this is a policy-budget proxy, not a deployed invocation count.

Shift and exploratory transfer. On 53 development slices, tri-band gating adds +2.08 pp weighted accuracy (+0.23 pp ID, +7.44 pp OOD); 6/53 regress (Figure 2D). Task-adapted pilots give ProntoQA +0.12 pp, GSM8K up to +1.28 pp [4], StrategyQA +0.62 pp (95% refinement), ProofWriter +0.00 pp, and MATH-500 +6.00 pp ($N{=}50$, 92% refinement). StrategyQA/ProofWriter use oracle-like depth; hence these pilots show feasibility, not task-independent generalization or selective-compute superiority.

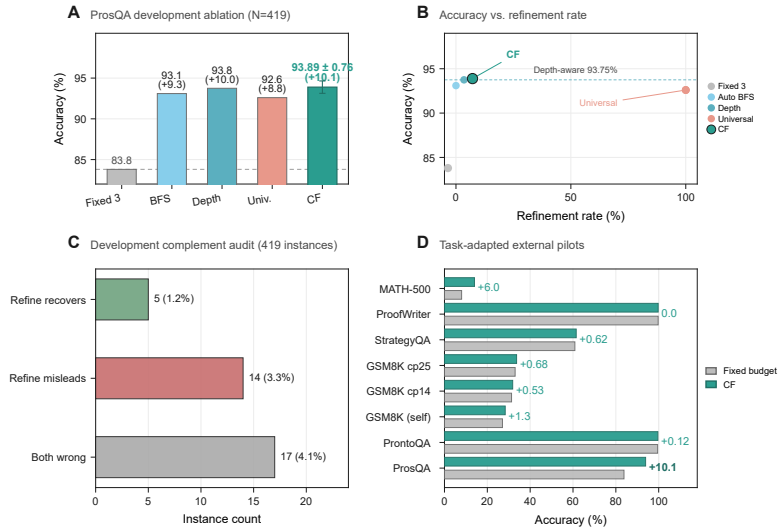


Figure 2: **ProsQA development ablation and exploratory external results. A**, Five-policy ablation (Fixed 3 83.8%; CF mean 93.89±0.76%, best seed 95.23%). **B**, Accuracy–refinement tradeoff. **C**, Path-union oracle ceiling 95.94%. **D**, Fixed-budget vs. task-adapted CF pilots; ProofWriter Δ=0.0 pp reflects a 99.75% ceiling.

5 Conclusion

Separating depth mismatch from commitment error explains why route-then-gate control can outperform uniform refinement. With a 106K-parameter stop head on frozen Coconut weights, CF reaches 93.89±0.76% on the ProsQA development set (best 95.23%) while refining 7.2%. The supported claim is an accuracy–policy-budget improvement among methods sharing this frozen base; commercial APIs are direct-answer references (Table 1).

Limitations. The stop head is fitted on a subset of the 419 questions, and the full set is reused for policy and threshold selection; consequently, the reported ProsQA values are results of actual experiments on the development set, rather than unbiased held-out test results, and require confirmation on an untouched test set. BFS routing is task-specific; external pilots recalibrate components, two use oracle-like depth, and MATH-500 has only 50 examples. The 53 slices informed development; 17/419 main examples fail on both paths; and commercial results are not matched by scale, sample, or compute. Future work should freeze a task-agnostic controller, hold out whole tasks, and add equal-compute open-weight baselines.

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